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A Proofs

This section contains additional proofs. The proof of Proposition 1 is in Section A.1. Proposition 3 is proved in Section A.2.

A.1 Proof of Proposition 1

Proposition 1 follows from the more general result that we state below. The statement and proof are similar to Gretton et al. (2012, Theorem 8) but do not assume $m = n$. Note that we recover Gretton et al. (2012, Theorem 8) in the case that $m = n$. We prove Proposition 1 afterwards.

Proposition 4. *Let $\mathbb{P}$, $\mathbb{Q}$, $\hat{\mathbb{P}}_m$, $\hat{\mathbb{Q}}_n$ be defined as in the main text, assume $0 \leq k(x, y) \leq K$ for all $x, y \in \mathcal{X}$, $\mathbb{P} = \mathbb{Q}$, and $t > 0$. Then*

$$P\left(\mathrm{MMD}\left(\hat{\mathbb{P}}_m, \hat{\mathbb{Q}}_n\right) - \left(\frac{K}{m} + \frac{K}{n}\right)^{\frac{1}{2}} \geq t\right) \leq e^{-\frac{t^2 mn}{2K(m+n)}}.$$

Proof. First, we bound the difference of $\mathrm{MMD}\left(\hat{\mathbb{P}}_m, \hat{\mathbb{Q}}_n\right)$ to its expected value. Changing a single one of either x_i or y_j in this function results in changes of at most $2\sqrt{K}/m$, and $2\sqrt{K}/n$, giving

$$\sum_{i=1}^{n+m} c_i^2 = 4K\frac{n+m}{nm}.$$

We now apply the bounded differences inequality (recalled in Theorem 1) to obtain

$$P\Big(\mathrm{MMD}\left(\hat{\mathbb{P}}_m, \hat{\mathbb{Q}}_n\right) - \mathbb{E}\,\mathrm{MMD}\left(\hat{\mathbb{P}}_m, \hat{\mathbb{Q}}_n\right) \geq t\Big) \leq e^{-\frac{t^2 mn}{2K(m+n)}}.$$

The last step is to bound the expectation, which yields

$$\begin{aligned}
\mathbb{E}\,\mathrm{MMD}\left(\hat{\mathbb{P}}_m, \hat{\mathbb{Q}}_n\right) &= \mathbb{E}\left(\frac{1}{m^2}\sum_{i,j=1}^{m} k(x_i, x_j) + \frac{1}{n^2}\sum_{i,j=1}^{n} k(y_i, y_j) - \frac{1}{mn}\sum_{i,j=1}^{m,n} k(x_i, y_j) - \frac{1}{mn}\sum_{j,i=1}^{n,m} k(y_j, x_i)\right)^{\frac{1}{2}} \\
&\leq \left(\frac{1}{m}\mathbb{E}k(X,X) + \frac{1}{n}\mathbb{E}k(Y,Y) + \frac{1}{m}(m-1)\mathbb{E}k(X,Y) + \frac{1}{n}(n-1)\mathbb{E}k(Y,X) - 2\mathbb{E}k(X,Y)\right)^{\frac{1}{2}} \\
&= \left(\frac{1}{m}\mathbb{E}k(X,X) + \frac{1}{n}\mathbb{E}k(Y,Y) - \frac{1}{m}\mathbb{E}k(X,Y) - \frac{1}{n}\mathbb{E}k(X,Y)\right)^{\frac{1}{2}} \\
&= \left(\frac{1}{m}\mathbb{E}\left[k(X,X) - k(X,Y)\right] + \frac{1}{n}\mathbb{E}\left[k(X,X) - k(X,Y)\right]\right)^{\frac{1}{2}} \leq \left(\frac{K}{m} + \frac{K}{n}\right)^{\frac{1}{2}}.
\end{aligned}$$

Inserting this into the previous inequality, we obtain the stated result. □